



Regular Article

Endothelial cell-selective adhesion molecule regulates albuminuria in diabetic nephropathy

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ABSTRACT

Microalbuminuria is a primary manifestation of diabetic nephropathy. Endothelial cell-selective adhesion molecule (ESAM) is a new member of the immunoglobulin superfamily which is selectively expressed by vascular endothelial cells. Although ESAM mediates homophilic interaction between endothelial cells, the role of ESAM in glomerular permeability remains unknown. We examined the expression and function of ESAM in the high glucose-induced microangiopathy in the kidney. ESAM was highly expressed in the glomerular endothelial cells, and the level was significantly reduced in the streptozotocin-induced diabetic mice. Stimulation of cultured endothelial cells with high glucose (35 mmol/l) resulted in a significant decrease in the ESAM expression compared to normal glucose (5.5 mmol/l). *In vitro* permeability assays revealed that albumin diffusion across endothelial monolayers was significantly increased when ESAM was knocked down by siRNA, suggesting that ESAM regulates vascular permeability of the glomeruli. To confirm these results *in vivo*, albuminuria was assessed using ESAM^{−/−} mice. Urinary albumin to creatinine ratio in ESAM^{−/−} mice was significantly higher than in ESAM^{+/+} mice. Transmission electron microscopy revealed that glomerular endothelial fenestration was decreased and endothelial tight junction was irregular and relatively wider in ESAM^{−/−} mice than in ESAM^{+/+} mice. In conclusion, hyperglycemia down-regulates ESAM and increases glomerular endothelial permeability. Thus, ESAM may regulate albumin extravasation at the glomeruli and play a role in the initiation of diabetic nephropathy.

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Diabetic nephropathy is a major complication of diabetes and a leading cause of end-stage renal failure. Endothelial dysfunction is a primary and fundamental step in the pathogenesis of diabetic nephropathy, but the role of the glomerular endothelium in the initiation of diabetic nephropathy has been incompletely understood. Albuminuria is a cardinal feature of a variety of glomerular diseases including diabetic nephropathy. Selective leakage of plasma albumin into the Bowman's cavity is mainly regulated by the permeability of the glomerular filtration barrier, which is comprised of three layers: a fenestrated endothelial layer, the glomerular basement membrane (GBM), and podocyte foot processes connected by a slit diaphragm (Ronco, 2007). Traditionally, it was considered that the GBM and podocyte foot processes play a major role in the barrier function (Sever et al., 2007; Jarad et al. 2006) and the glomerular endothelium contributes little to the permselectivity of the glomerular capillary wall. However, it has been recently demonstrated that the glomerular endothelium may play a significant role in the regulation of the glomerular filtration (Ballermann, 2007; Ballermann and Stan, 2007). Thus, the role of glomerular endothelial cells in the genesis of albuminuria is still controversial.

Endothelial cell-selective adhesion molecule (ESAM) is a novel endothelial cell surface protein which belongs to the immunoglobulin superfamily (Hirata et al., 2001). Previous studies have demonstrated that ESAM expression is restricted to endothelial cells in the embryonic and adult vasculature (Hirata et al., 2001; Nasdala et al., 2002). ESAM has been shown to mediate homophilic and calcium-independent adhesion of the expressing cells (Hirata et al., 2001) and regulate the vascular permeability (Ishida et al., 2002; Wegmann et al., 2006). In addition, ESAM is expressed at the glomeruli and co-localized with endothelial markers (Nasdala et al., 2002). These lines of evidence suggest that ESAM may play a pivotal role in the genesis of glomerular function, although the role of ESAM in the glomerular filtration barrier has not been reported. To investigate the pathogenesis of albuminuria in diabetic nephropathy, in this study, we aimed to examine the expression of ESAM under hyperglycemic conditions, and explore the functional role of ESAM in a murine model of hyperglycemia.

Materials and methods

Cell culture, transfection, and siRNA experiments

The PY4.1 (a mouse endothelial cell line) was a gift from Dr. T. Quertermous (Stanford University, Stanford, CA). PY4.1 cells stably

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expressing FLAG-tagged human ESAM (ESAM-PY4.1) were obtained by transfection with the *p3xFLAG-CMV9* expression vector (Sigma-Aldrich, St. Louis, MO) containing human ESAM cDNA using the Lipofectamine 2000 reagent (Invitrogen) and by selection with G418. Human umbilical vein endothelial cells (HUVEC) were purchased from the American Type Culture Collection (Manassas, VA). HUVEC at passages 4–10 was cultured on gelatin-coated culture plates according to the manufacture's recommendations. For the stimulation assay, subconfluent HUVEC cultured on 6-well plates were treated for 24 h with fresh complete medium containing 5.5 mmol/l D-glucose, 35 mmol/l D-glucose, or and 35 mmol/l D-mannitol (Wako Pure Chemicals, Tokyo, Japan). The siRNA-mediated knockdown of ESAM was performed using a siGENOME SMARTpool specific for human ESAM (Dharmacon, Lafayette, CO). HUVEC was transfected with the ESAM siRNA or Stealth RNAi negative control using the Lipofectamine RNAiMAX reagent (Invitrogen, Carlsbad, CA) according to the manufacture's protocol. After incubation for 24–72 h, cells were subjected to the *in vitro* permeability assay.

Permeability assay

For the detection of endothelial permeability, albumin diffusion across endothelial monolayers was measured as previously described (Patterson et al., 1992; Bove et al., 2001). Briefly, after cell culture inserts (6.5-mm diameter, 0.4- μ m pore size; Corning-Costar, Acton, MA) were coated with gelatin, HUVEC was seeded on the inserts at a density of 2×10^5 cells/insert, transfected with siRNA prior to the stimulation with high glucose. We verified by phase-contrast microscopy that endothelial monolayer was formed in each well. The PY4.1 cells were seeded on the inserts at a density of 2×10^5 cells/insert, and then the permeability assays were performed when the cells reached confluency. The permeability was characterized by the clearance rate of albumin labeled with Evans blue dye (EBD) (Patterson et al., 1992; Bove et al., 2001). Hanks' balanced salt solution (HBSS) containing 4.0% BSA and 20 mmol/l N-2-hydroxyethylpiperazine-N'-2-ethanesulfonic acid buffer was used on both sides of the monolayer. The albumin was labeled with EBD by addition of stock EBD solution (20 mg/ml in HBSS) to the upper compartment to the final concentration of 0.67 mg/ml. To determine the efficiency of the binding of EBD to albumin, the dye at 0.67 mg/ml in 4% albumin was placed in the dialysis tube (Spectra-Por 15000-mol wt exclusion, Spectrum Medical, Los Angeles, CA). We confirmed that more than 99% of the EBD bound to albumin. The samples (100 μ l) were taken from the lower compartment after 60 min incubation. The absorbance (620 nm) of the samples was measured in a microplate spectrophotometer.

Induction of diabetes in mice by streptozotocin injection

All animal experiments were conducted according to the Guidelines for Animal Experiments at Kobe University Graduate School of Medicine. Adequate measures were taken to minimize pain or discomfort, and that the experiments were conducted in accordance with international standards on animal welfare. We generated mice

Table 1
Characteristics of control and streptozotocin-induced diabetic mice

	Control mice	STZ-injected mice
Blood glucose before injection (mg/ml)	177.0 $\pm$ 26.4	189.8 $\pm$ 42.1
Blood glucose 4 weeks after injection (mg/ml)	165.1 $\pm$ 24.8	545.2 $\pm$ 60.9*
Weight on the day of injection (g)	23.7 $\pm$ 1.2	23.5 $\pm$ 1.5
Weight 4 weeks after injection (g)	26.7 $\pm$ 1.5	21.6 $\pm$ 2.3*

Mice received a single injection of streptozotocin (STZ, 200 mg/kg, i.p.) or vehicle citrate buffer (control). The blood glucose level and body weight of the mice were measured before and 4-weeks after the injection of STZ.

* $p < 0.001$ compared with control.

Table 2
Primers for quantitative RT-PCR

Gene	Forward primer (5'-3')	Reverse primer (5'-3')
Mouse ESAM	tgtcaatgtgcagaatgatgaagg	tgcagttcagggtcacattgg
Mouse GAPDH	tgtgtccgtcgtggatctga	ttgctgttgaaagtcgaggag
Human ESAM	ggagtaagcccgctgtccaa	cagacatagactccagccatggaa
Human GAPDH	ggctccaaggagtaagacc	agggtgtcatatggcaactg

lacking ESAM (ESAM $^{-/-}$) as previously reported (Ishida et al., 2003). A single high-dose streptozotocin (STZ) regimen was used to induce pancreatic islet cell destruction and persistent hyperglycemia, as previously described (Anitha et al., 2006; Alp et al., 2003). STZ (Wako Pure Chemicals) was freshly dissolved in 0.05 mol/l citrate buffer (pH 4.5) and used within 10 min. Wild-type C57BL/6 mice of 8–12 weeks old received a single injection of STZ (200 mg/kg, i.p.) or vehicle citrate buffer (control). The blood glucose level and body weight of the mice were measured prior to injection as well as 72 h and 4 weeks after injection of STZ. The blood glucose levels are shown in Table 1 (72-hour measurement not shown). Hyperglycemia was defined as a random blood glucose level of more than 20 mmol/l for more than 3 weeks after the STZ injection.

Urinalysis

For urinalysis, male ESAM $^{-/-}$ mice of 8 weeks old and age-matched wild-type C57BL/6 (ESAM $^{+/+}$) mice were used. Mice received a single injection of STZ (200 mg/kg, i.p.) or vehicle citrate buffer (control), and 4 weeks later, urinary samples were collected for 24 h using a metabolic cage (CLEA Japan, Shizuoka, Japan). Urinary albumin and creatinine concentrations were measured by the standard immunoassay to calculate albumin/creatinine ratio (Alb/Cr).

Real-time polymerase chain reaction

Total RNA was extracted from mouse tissues and cultured cells by the guanidine thiocyanate method using the TRIZOL reagent (Invitrogen) according to the manufacture's protocol. For real-time polymerase chain reactions (PCR), total RNA (500 ng) was reverse transcribed utilizing standard methodology. PCR primers for mouse and human ESAM and GAPDH were purchased from Takara-Bio Perfect Real Time Support System (Shiga, Japan), and the sequences are shown in Table 2. Quantitative real-time PCR was performed with an ABI PRISM 7500 Sequence Detector (Applied Biosystems, Foster City, CA) according to the comparative threshold cycle method following manufacture's protocol (SYBER Premix Ex TaqTM, Takara-Bio). The PCR reaction was performed in a 20 μ l total reaction mixture in 96-well plates containing; 2 μ l of cDNA as template, 10 μ l of SYBER Premix Ex Taq, 0.4 μ l of ROX dye (Takara-Bio), and 0.5 μ l of 10 μ mol/l each of the forward and reverse primers. Thermal cycler conditions were 95 $^{\circ}$ C for 10 s, and 40 cycles of 95 $^{\circ}$ C for 5 s and 60 $^{\circ}$ C for 34 s, followed by 95 $^{\circ}$ C for 15 s, 60 $^{\circ}$ C 1 min, and 95 $^{\circ}$ C 15 s.

Immunoblotting

Kidney tissues were excised from mice and homogenized in a lysis buffer (10 mmol/l Tris/HCl (pH 7.4), 150 mmol/l NaCl, 2 mmol/l CaCl₂, 1% Nonidet P-40, 1% Triton X-100, 1 mmol/l phenylmethylsulfonyl fluoride, 40 units/ml aprotinin, and 15 μ g/ml leupeptin). Western blotting was performed using anti-ESAM polyclonal antibody (R & D Systems, Minneapolis, MN), anti-VE-cadherin antibody (Santa-Cruz Biotechnology, Santa Cruz, CA), or anti- β -actin monoclonal antibody (Sigma-Aldrich, St. Louis, MO). The relative densities of the acquired bands were determined by using the ImageJ software version 1.38 from the National Institutes of Health (available on the World Wide Web).

Histological analysis

For immunohistochemistry, mouse kidneys were fixed with 10% formalin and then embedded in Tissue Tek OCT compounds (Tissue-Tek, Torrance, CA), snap frozen, and stored at -80°C . Sections of 8–12 μm were cut on a freezing microtome, mounted on slides. After the reduction of endogenous peroxidase activity with 0.1% hydrogen peroxide for 15 min at room temperature, nonspecific binding was blocked with 2% bovine serum albumin in PBS for 30 min. The tissue sections were incubated for 1 h with anti-ESAM antibody (R & D Systems), or von Willebrand factor antibody (ABCAM, Tokyo, Japan) diluted in PBS, followed by incubation with N-Histofine[®] Simple Stain Mouse MAX PO (Nichirei Bioscience, Tokyo, Japan).

Light microscopy and transmission electron microscopy

Kidney tissues were fixed in 10% formalin, and embedded in paraffin. The section was sliced with 5- μm thickness, dewaxed, and stained with PAM. Glomerular size, tuft area, and mesangial matrix area were evaluated as previously reported (Okada et al., 2003). The mesangial matrix index represented the ratio of mesangial matrix area divided by the tuft area. Five randomly selected glomeruli in the cortex were examined in each group.

Tissues for electron microscopy were fixed in 2% glutaraldehyde, followed by 0.1 mol/l phosphate buffer. After hydrated by graded ethanol, specimens were infiltrated and embedded with epoxy resin. Ultra-thin slices were cut with a diamond knife and then examined. For morphological assessment of glomerular tight junctions, a total of 12 tight junctions of vascular endothelial cells in the kidney were examined (Schbert et al., 2002) in each genotype. We measured maximal and minimal widths of the tight junction, which is referred to as TJ_{max} and TJ_{min} , respectively, in each sample.

Statistics

The results are expressed as the mean $\pm$ SEM. The significance of the differences among the experimental groups was determined by a one-way analysis of variance followed by the Bonferroni test for multiple comparisons. The level of statistical significance was set at $p < 0.05$.

Results

High glucose decreases ESAM expression in vascular endothelial cells

We examined if ESAM expression in vascular endothelial cells is regulated by glucose concentrations. Real-time PCR revealed that incubation of HUVEC with the high concentration of glucose (35 mmol/l) resulted in a significant decrease in the ESAM expression compared to that with normal glucose (5.5 mmol/l) (Fig. 1A). Because high glucose condition sometimes affects gene expression via an osmotic effect (Taki et al., 1996), we treated HUVEC with 35 mmol/l mannitol and evaluated the ESAM expression. The mannitol treatment did not affect the ESAM expression (Fig. 1A). Immunoblot analysis confirmed that the treatment of HUVEC with high glucose for 72 h decreased ESAM expression ($p < 0.05$, Fig. 1B). These findings suggest that high glucose decreased ESAM expression in vascular endothelial cells not through the osmotic effect.

Decrease of ESAM expression increases endothelial permeability

Because ESAM has been reported to regulate vascular permeability (Ishida et al., 2002; Wegmann et al., 2006), we examined how alteration of ESAM expression affects cellular permeability. First, we analyzed the endothelial permeability in the presence or absence of high glucose stimulation. *In vitro* permeability assays revealed that

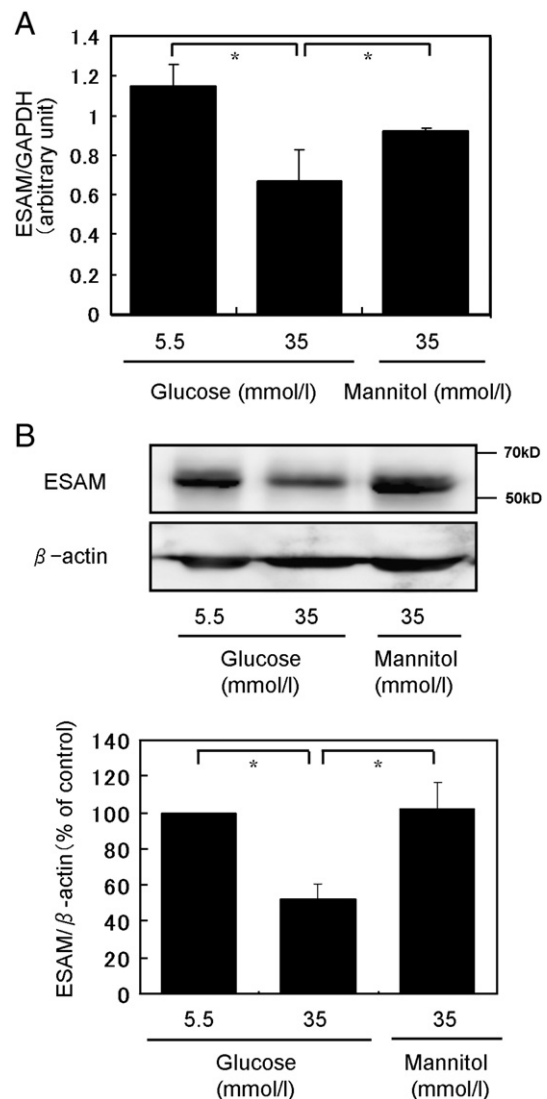


Fig. 1. Regulation of ESAM expression by high glucose stimulation. (A) Subconfluent HUVECs cultured in 6-well plates were treated with fresh complete medium containing low glucose (5.5 mmol/l glucose), high D-glucose (35 mmol/l), or 35 mmol/l D-mannitol. After 24 h incubation, total RNA samples were extracted and ESAM expression was analyzed by real-time PCR. Bars represent means $\pm$ SE. $*p < 0.05$ ($n = 6$ per group). (B) Subconfluent HUVECs cultured in 60-mm plates were treated with the medium containing low glucose, high glucose, or mannitol. After 72 h incubation, protein samples were extracted and analyzed by immunoblotting. Representative immunoblotting image is shown (upper). Densitometric quantification of 4 independent experiments is also shown (lower). Values are expressed as percent of control (low glucose group). Bars represent means $\pm$ SE. $*p < 0.05$ ($n = 4$).

the treatment of HUVEC with high glucose resulted in significant increase of endothelial permeability (Fig. 2A), suggesting that the reduction of ESAM expression by high glucose directly increased the endothelial permeability.

Because high glucose may affect expressions of various molecules that regulate endothelial permeability, we tried to selectively manipulate ESAM expression and examined the association between ESAM expression and cellular permeability. First, we decreased ESAM expression in HUVEC by transfection of siRNA (Fig. 2B, upper panel) and evaluated endothelial permeability. *In vitro* permeability assays revealed that the ESAM-knockdown resulted in a significant increase in endothelial permeability, compared with the control siRNA treatment ($p < 0.05$ vs. control; $n = 6$, Fig. 2B, lower panel). Next, the effect of ESAM overexpression was evaluated using endothelial PY4.1 cells which stably express ESAM. The ESAM-overexpressing PY4.1 cells

exhibited a significant decrease in BSA clearance compared to control PY4.1 cells ($p < 0.05$, $n = 6$, Fig. 2C). Further, we checked whether stable overexpression of ESAM would reverse the increased permeability induced by high glucose treatment. Notably, PY4.1 cells that constitutively overexpress ESAM did not show the increase in permeability in response to the high glucose treatment ($p < 0.05$, $n = 6$, Fig. 2C). Therefore, there was an inverse gene-dosage effect between ESAM expression and cellular permeability.

ESAM expression in the kidney is decreased in diabetic mice

We examined the expression of ESAM in the nephron of wild-type C57Bl/6 mice. Immunohistochemical analysis revealed that ESAM was selectively expressed in the glomerular endothelial cells (Fig. 3A), which is consistent with a previous study (Nasdala et al. 2002). Next, we examined if hyperglycemia affects the level of ESAM *in vivo*. STZ-induced diabetes was generated in C57Bl/6 mice, as shown in Table 1. Real-time PCR revealed that ESAM mRNA expression was significantly reduced in the kidney from STZ-induced diabetic mice compared to that from vehicle-treated mice ($p < 0.05$, Fig. 3B). Also, western blotting revealed that ESAM protein levels were decreased in the kidney from STZ-induced diabetic mice compared to those from vehicle-treated mice ($p < 0.05$, Fig. 3C). In order to exclude the possibility that this reduction reflects the change in cellular composition in the kidney, we examined the level of other endothelial adhesion molecules such as VE-cadherin, but found no change in the expression of VE-cadherin (Fig. 3C).

ESAM $^{-/-}$ mice exhibited an increase in urinary albumin excretion

Because our cell culture experiments demonstrated that ESAM downregulation by hyperglycemia resulted in an increase in endothelial permeability, we examined if ESAM downregulation in hyperglycemic mice was associated with albumin secretion in the urine. Urinary Alb/Cr was about 1.6-fold higher in ESAM $^{-/-}$ mice than that in ESAM $^{+/+}$ mice ($p < 0.05$, Fig. 4). On the other hand, that in STZ-injected mice was 2.3-fold higher than control ($p < 0.05$, Fig. 4). These findings indicate that the increase of urinary Alb/Cr by high glucose treatment was, at least partially, mediated by the decrease of ESAM expression levels. Total protein concentrations in the urine did not differ between the mouse groups (data not shown).

Fenestration of glomerular capillary endothelial cells was decreased in ESAM $^{-/-}$ mice

To further clarify the mechanism for the increased albumin secretion in ESAM $^{-/-}$ mice, we performed the histological analysis of the kidney from ESAM $^{-/-}$ and ESAM $^{+/+}$ mice. In the light microscopy, there was no overt morphologic change observed in the kidney between ESAM $^{+/+}$ and ESAM $^{-/-}$ mice. PAM stain revealed no significant difference in the glomerular size, tuft area, and

mesangial matrix index (Fig. 5A). However, electron microscopy revealed that the number of fenestration of glomerular capillary endothelial cells was significantly decreased in ESAM $^{-/-}$ mice compared with in ESAM $^{+/+}$ mice (Fig. 5B). No structural change was observed in podocytes or GBM, such as podocyte effacement, mesangial expansion, and GBM denudation, which are typically observed in diabetic nephropathy, between ESAM $^{+/+}$ mice and ESAM $^{-/-}$ mice. Therefore, ESAM deficiency selectively decreased the fenestration of glomerular capillary endothelial cells.

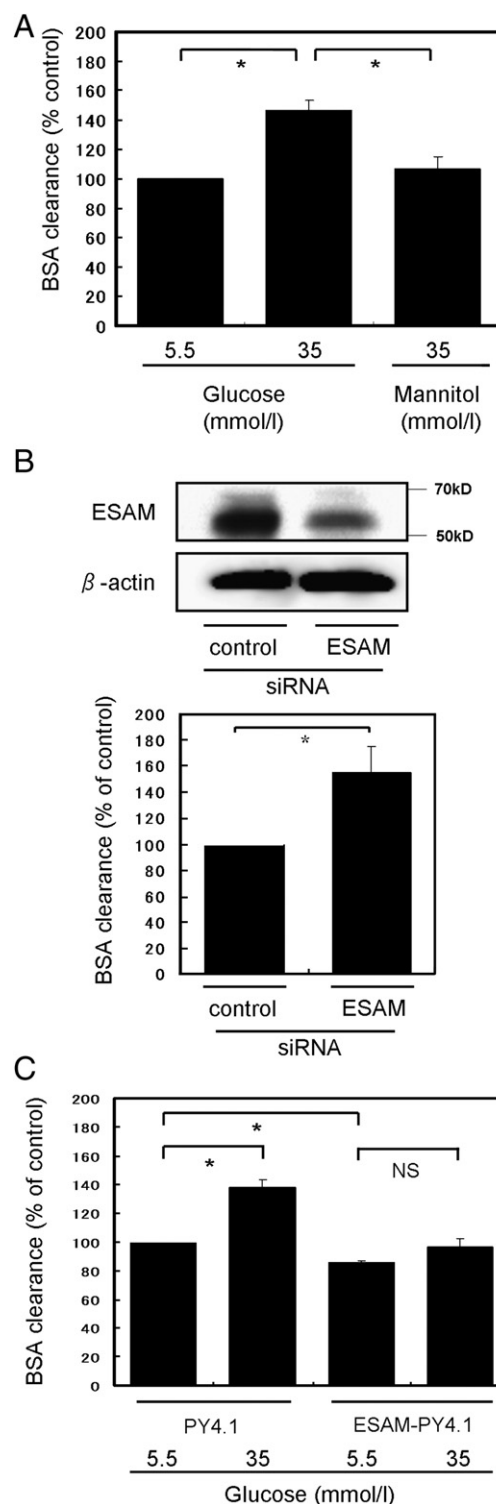


Fig. 2. Involvement of ESAM in the albumin clearance. (A) HUVEC cells were seeded on the upper well of Transwell® system and incubated with fresh complete medium containing low glucose (5.5 mmol/l glucose), high D-glucose (35 mmol/l), or 35 mmol/l D-mannitol. For the permeability assay, after 72 h incubation, BSA bound with Evans-blue dye was added onto the upper well, and the absorbance (620 nm) of the medium in the lower well was measured. Values are expressed as percent of control (low glucose group). Bars represent means $\pm$ SE. * $p < 0.05$ ($n = 9$ per group). (B) HUVEC was seeded on the upper well of Transwell® system, and transfected either with control siRNAs (control) or with siRNAs directed against ESAM (ESAM). Forty-eight hours later, cell lysates were subjected to immunoblotting for ESAM and β -actin (upper panel). BSA clearance between the endothelial monolayer was analyzed. Values are expressed as percent of control. Bars represent means $\pm$ SE. * $p < 0.05$ vs. control ($n = 6$). (C) Control PY4.1 cells and ESAM-overexpressing PY4.1 (ESAM-PY4.1) were seeded and incubated with low or high glucose medium for 72 h, and BSA clearance was analyzed. Values are expressed as percent of control (low glucose-treated PY4.1). Bars represent means $\pm$ SE. * $p < 0.05$ vs. control ($n = 6$), NS = not significant.

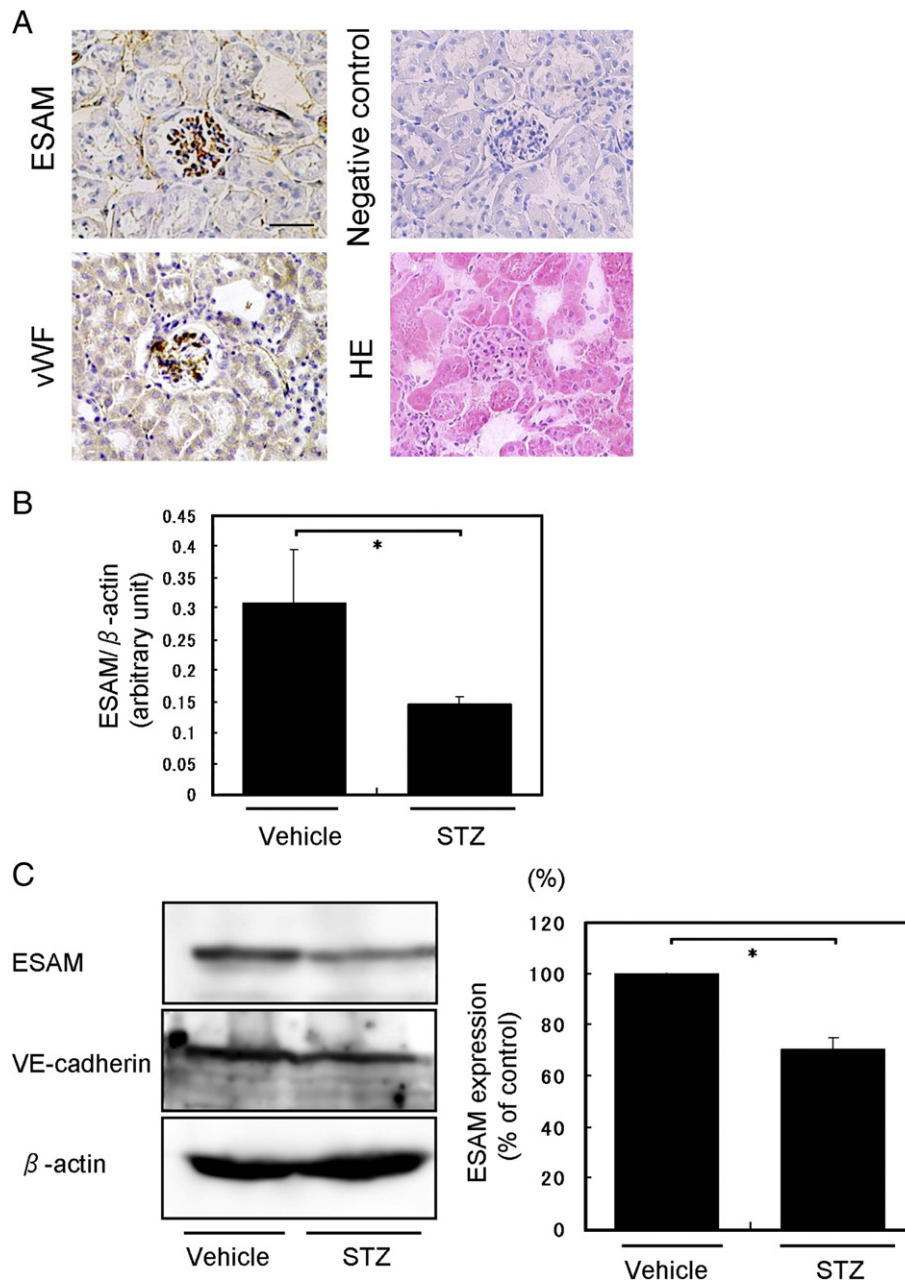


Fig. 3. ESAM expression in the mouse kidney. (A) Immunohistochemistry for ESAM expression at the glomeruli. Frozen sections of the mouse kidney were stained with anti-ESAM polyclonal antibodies or von Willebrand factor (vWF). ESAM expression was detected in the glomeruli and colocalized with vWF. The bar represents 50 μ m. HE represents hematoxylin and eosin staining. (B) ESAM levels in the kidney of STZ-injected mice. Wild-type C57BL/6 mice were injected with STZ (200 mg/kg, i.p.). Four weeks after the STZ injection, RNA samples were extracted from the kidney and ESAM expression was examined by real-time PCR. Bars represent means $\pm$ SE. * p < 0.05 vs. control (n = 5 per group). (C) Immunoblotting for ESAM and VE-cadherin in the kidney of STZ-injected mice, confirming the decrease in the ESAM expression in the diabetic mice. Densitometric quantification of 5 independent experiments is also shown (right). Values are expressed as percent of control (vehicle treatment). Bars represent means $\pm$ SE. * p < 0.05 vs. control (n = 5 per group).

ESAM^{-/-} mice exhibited expanded tight junctions between vascular endothelial cells

We then examined whether ESAM deficiency results in a morphological change in tight junctions. Electron microscopic examination of glomerular endothelial cells in ESAM^{-/-} mice showed extended openings along the endothelial cell junctions when compared with wild-type C57BL/6 mice (Fig. 6A). The maximal width of tight junctions in ESAM^{-/-} mice was significantly larger than that of ESAM^{+/+} mice, indicating that open and expanded tight junctions were evident in ESAM^{-/-} mice. Besides, the ratio between maximal and minimal tight junction width was also significantly larger in ESAM^{-/-} mice than ESAM^{+/+} mice (p < 0.05, Fig. 6B), which suggests that the width of

tight junction was irregular in ESAM^{-/-} mice. These findings suggest that the increased endothelial permeability by ESAM deficiency may be associated with the morphological change of tight junctions between vascular endothelial cells.

Discussion

ESAM is abundantly expressed at the tight junction in vascular endothelial cells, and mediates homophilic adhesion between endothelial cells (Hirata et al., 2001; Nasdala et al., 2002). Previously, we reported for the first time that ESAM deficiency in endothelial cells directly increases vascular permeability (Ishida et al., 2002). Recently, Wegmann and colleagues have further documented the role of ESAM

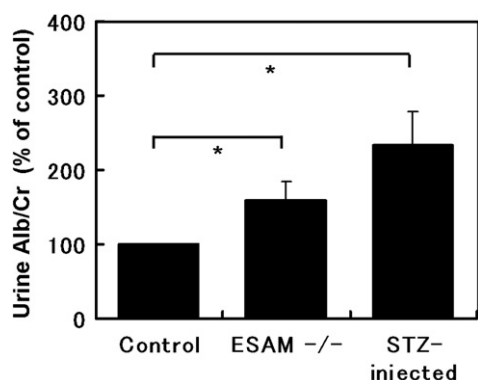


Fig. 4. ESAM deficiency increased urine albumin secretion. Urine albumin concentration in ESAM^{-/-} mice significantly increased compared with ESAM^{+/+} mice. Values are expressed as percent of control. Bars represent means $\pm$ SE. * $p < 0.05$ vs. ESAM^{+/+} ($n = 8$ per group).

in VEGF-induced vascular permeability (Wegmann et al., 2006). Moreover, ESAM has been shown to interact with intracellular molecules such as MAGI-1, suggesting that ESAM can transduce the intracellular signals in endothelial cells, and thereby indirectly modulate interendothelial interactions (Wegmann et al., 2004). From these findings, it was reasonable to assume that the reduction of ESAM expression could attenuate the mechanical adhesion and

directly make the endothelial cell contact loose. The present study has directly demonstrated that ESAM expression was inversely correlated with the cellular permeability. Moreover, tight junctions between vascular endothelial cells were frequently irregular and wider in ESAM^{-/-} mice. These morphological changes in the tight junction are similar to the effect of vascular endothelial growth factor (VEGF) on endothelial tight junctions (Behzadian et al., 2003). VEGF is one of the most potent permeabilizing factors of the vascular endothelium. In particular, it has been demonstrated that VEGF is implicated in the genesis of albuminuria in diabetes. VEGF levels are elevated in the diabetic kidney, and its urinary excretion is increased. Blockade of VEGF signaling ameliorates albuminuria in diabetes (Sung et al., 2006). These lines of evidence serve to lend credence to the assumption that VEGF-induced morphological changes in endothelial tight junctions may play some role in albuminuria in diabetic states through increasing endothelial permeability. In this context, morphological changes in the tight junction induced by ESAM downregulation are likely related to the increase in glomerular endothelial permeability. ESAM expression was decreased by the hyperglycemic stimulation both *in vivo* and *in vitro*, which was concomitant with the increase in trans-endothelial permeability of albumin. These findings indicate that ESAM regulates the glomerular barrier function and plays an important role in the pathophysiology of albuminuria in the early stage of diabetic nephropathy.

We found the decrease of ESAM expression in diabetic mice during rather early phase (4 weeks). When we evaluated the ESAM

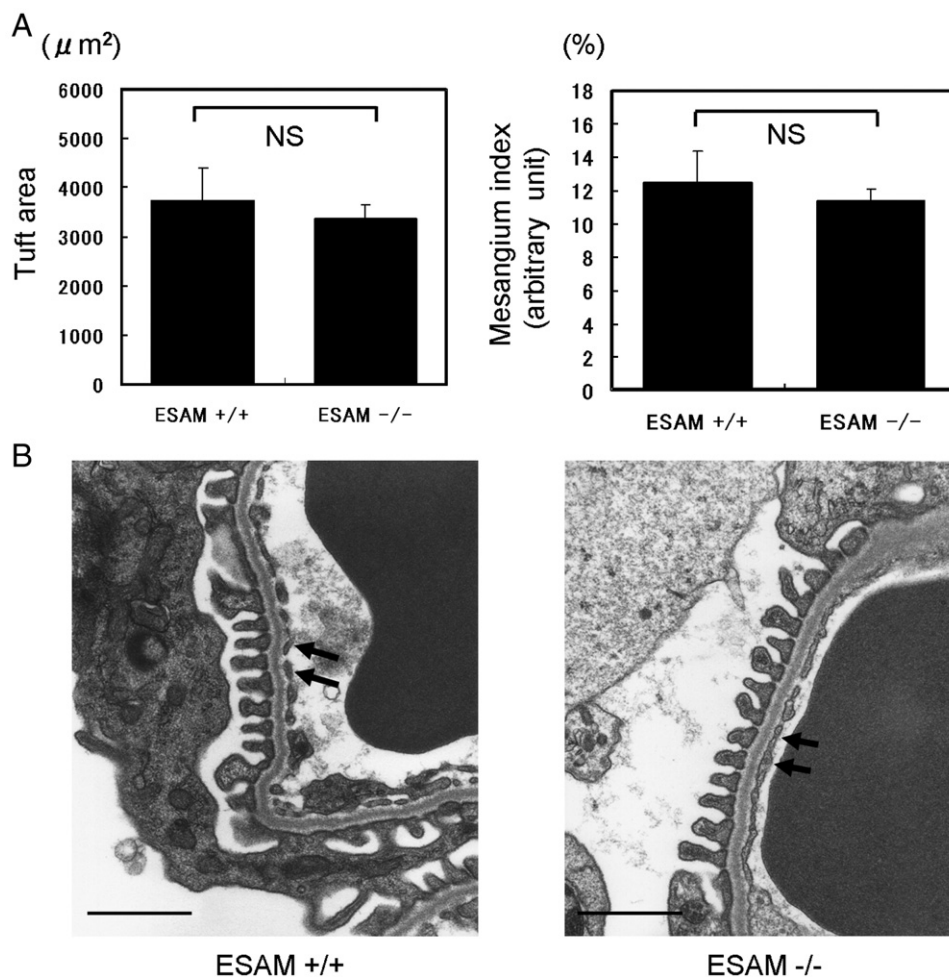


Fig. 5. Cellular composition and morphology in the glomeruli in ESAM^{-/-} mice. (A) Tuft area (left) and mesangial index (right) in the glomeruli. No difference was observed both in tuft area and mesangial matrix index between ESAM^{+/+} and ESAM^{-/-} mice. Bars represent means $\pm$ SE ($n = 5$ per group). NS = not significant. (B) Transmission electron microscopy revealed the decreased glomerular endothelial fenestration in ESAM^{-/-} mice. Arrows denote areas of glomerular endothelial cells. Original magnification, $\times 15,000$. Scale bar, 500 nm.

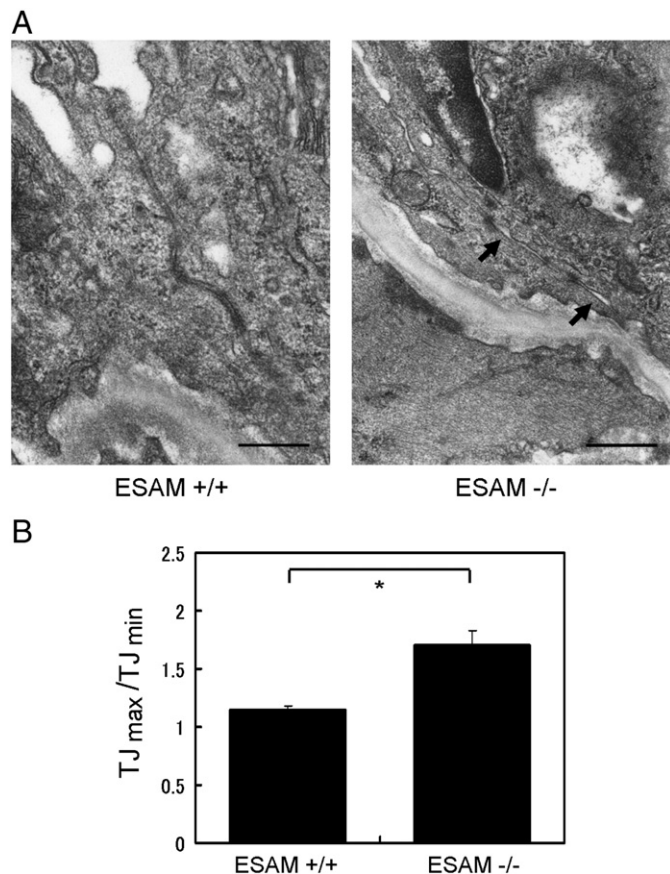


Fig. 6. Morphological examination of tight junctions in vascular endothelial cells. (A) Transmission electron microscopy revealed the expanded tight junctions of vascular endothelial cells in *ESAM*^{-/-} mice. Arrows denote areas of morphologically open inter-endothelial cell tight junctions. Original magnification, $\times 30,000$. Scale bar, 200 nm. (B) Quantification of the ratio between maximal and minimal tight junction width. TJ_{max} and TJ_{min} represent maximal and minimal width of the tight junction. Bars represent means $\pm$ SE. * $p < 0.05$ vs. *ESAM*^{+/+} ($n = 12$ per group).

expression at 24 weeks when advanced nephropathy is developed, we did not find any difference in *ESAM* expression (*data not shown*). On the other hand, total protein concentrations did not differ between the mouse groups throughout the observation period. These findings are likely to correlate with the fact that micro-albuminuria is the earliest manifestation of diabetic nephropathy. Thus, the decrease of *ESAM* expression in the diabetic mice may be relevant to the albuminuria in the early stage of diabetic nephropathy in humans.

The present study has shown that the number of fenestration in the glomerular capillary endothelial cells was decreased in *ESAM*^{-/-} mice compared with in *ESAM*^{+/+} mice, which is associated with the increase in urinary albumin excretion. The loss of fenestration in the glomerular endothelial cells is one of the pathological features of diabetic nephropathy, including podocyte effacement and GBM thickening (Dávila-Esqueda et al., 2005; Spencer et al., 2004; Toyoda et al 2007). It has been postulated that the loss of the glomerular endothelial fenestration may reflect an early process of endothelial damages and is associated with albumin leakage (Sugimoto et al., 2003; Toyoda et al 2007). However, the role of glomerular endothelial fenestration in a barrier function remains a matter of conjecture.

Previously, glomerular endothelial cells did not appear to regulate glomerular barrier functions. Today, however, it has been recognized that glomerular endothelial cells have a variety of direct and indirect properties in the genesis of albuminuria (Ballermann, 2007; Ballermann and Stan, 2007). Notably, there is a crosstalk between podocytes

and glomerular endothelial cells, which is critical for development of the glomerular filtration barrier and continue in the filtering glomeruli (Eremina et al., 2003). Toyoda et al. showed that podocyte detachment and diminished endothelial cell fenestration are related to classical diabetic nephropathy lesions and renal dysfunction in type 1 diabetic patients (Toyoda et al., 2007). Infusion of anti-VEGF antibody or sFlt-1 in mice leads to cellular damages in glomerular endothelial cells as well as epithelial cells (Sugimoto et al., 2003). Thus, the glomerular endothelial damage seems closely correlated to the epithelial damage, and the decreased endothelial fenestration *per se* may affect the integrity of the GBM and podocyte foot processes and can be a cause of albuminuria. Further studies are required to determine the role of *ESAM* in endothelial damage and in the loss of fenestration in the glomerular endothelial cells.

There is accumulating evidence that endothelial adhesion molecules play a pivotal role in the genesis of diabetic vascular complications (Okada et al., 2003; Rosenkranz et al., 1999; Xu et al., 2005). The expression of adhesion molecules such as vascular cell adhesion molecule-1 (VCAM-1), intercellular adhesion molecule-1 (ICAM-1) and E-selectin was increased by the high glucose stimulation (Taki et al., 1996; Quagliaro et al., 2005; Altannavch et al., 2004), in contrast that the expression of *ESAM* was decreased in response to high glucose. ICAM-1 deficient mice were resistant against renal injury after induction of diabetes, and this effect is mediated through suppressing macrophage infiltration into the glomeruli (Okada et al., 2003). On the other hand, p-selectin deficiency in mice markedly accelerated the glomerular damage with increased interstitial mononuclear leukocyte infiltrates although p-selectin is an adhesion receptor for neutrophils, monocytes, and T cells (Rosenkranz et al., 1999). As a result, p-selectin deficient mice showed the increased proteinuria. Furthermore, Xu et al. showed that p-cadherin was decreased in both diabetic glomeruli and high glucose stimulated podocytes (Xu et al., 2005). In the light of previous findings, *ESAM* may protect against the initiation or progression of diabetic nephropathy as is the case with p-selectin and p-cadherin. In this context, *ESAM* may play a distinct role from VCAM-1 and ICAM-1.

In summary, we have established evidence that high glucose down-regulates *ESAM* in the kidney and increases endothelial permeability both *in vitro* and *in vivo*. This effect was likely associated with the frequent expanded tight junctions in vascular endothelial cells together with the decrease in endothelial fenestration. The decreases of *ESAM* expression in the glomerular endothelium resulted in an increase in albumin concentration in the urine. Our findings suggest that *ESAM* is a novel regulator of albumin extravasation in the kidney and may play a role in the initiation of albuminuria in the hyperglycemic status.

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